

Implication of HSAT2 Repeat Sequence RNAs in ME/CFS and Long COVID: A Hypothesis

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July 2026

Prepared for the ISLC-PAIS Conference 2026

Human satellite 2 (HSAT2) is a pericentromeric repeat representing approximately 1.5% of the human genome. Under physiological conditions, it is transcriptionally silenced through DNA methylation and H3K9me3-dependent heterochromatin [1, 2]. In cancer, however, HSAT2 may become aberrantly expressed and promotes immune suppression through extracellular vesicle-mediated reprogramming of myeloid cells, leading notably to the expansion of myeloid-derived suppressor cells and immune exhaustion [3].

Here we propose that an analogous mechanism may contribute to the development of post-acute infection syndromes (PAIS), including Long COVID and ME/CFS. This model provides a potential explanation for the frequent observation that these disorders emerge following reactivation of latent herpesviruses. Indeed, HSAT2 RNAs are also induced upon herpesvirus replication. Herpesvirus infections are known to disrupt centromere organization and can induce transcription of pericentromeric satellite repeats, including HSAT2 [4–6]. Following repeated or closely spaced viral infections, cumulative changes in chromatin composition and loss of CpG methylation at pericentromeric loci may become sufficiently extensive to initiate a self-sustaining process of progressive epigenetic erosion [1, 7–10].

By analogy with cancer, we hypothesize that HSAT2 RNAs released in extracellular vesicles could propagate this altered epigenetic state, driving persistent immune dysregulation and immune exhaustion [3]. Rather than presenting definitive proof, this Perspective assembles the available evidence into a coherent mechanistic

framework linking the initiating trigger to downstream molecular events and clinical manifestations. It also generates specific, experimentally testable predictions that can now be tested using patient-derived samples [11, 12].

1 What HSAT2 Is and Why It Is Normally Silent

HSAT2 (human satellite 2) is a stretch of DNA repeated many times in the genome, occupying approximately 1.5% of all our DNA. It sits in pericentromeric regions: the DNA flanking each chromosome's central pinch point (the centromere, where the two arms meet). Repetitive sequences account for roughly half of the human genome—far more than the ~2% that encodes proteins—yet have historically been neglected.

In healthy cells, HSAT2 does not produce RNA. This silencing is maintained by two locks: (i) DNA methylation, in which small methyl groups are added to specific sites (CpG, cytosine–phosphate–guanine dinucleotides), and (ii) tightly packed chromatin called heterochromatin, reinforced by a chemical mark on the histone proteins around which DNA is wound (H3K9me3: histone H3 trimethylated at lysine 9). Together these prevent the cellular machinery from transcribing HSAT2 into RNA [1, 2].

2 What Happens in Cancer

In several cancers, HSAT2 escapes silencing and is transcribed into RNA. That RNA is packaged into extracellular vesicles (EVs)—tiny membrane-wrapped particles, roughly 100 nm wide, that cells shed into body fluids and travel via the bloodstream. When HSAT2-laden EVs are taken up by immature cells of the myeloid lineage (precursors of several innate immune cell types), they reprogram those cells to dampen immunity. Specifically, the EVs drive expansion of myeloid-derived suppressor cells (MDSCs)—a population of incompletely matured myeloid cells that suppress other immune cells and are typically expanded in disease.

MDSCs in turn suppress natural killer (NK) cells—rapid-response killers that destroy virus-infected and cancerous cells—and drive CD8⁺ T-cells (“killer” T-cells, critical for clearing infections) into exhaustion, a state in which they progressively lose function. This is mediated by the signalling molecules IL-10, IL-35, the enzyme IDO1, and TGF- β (transforming growth factor beta) [3]. Tumours expressing HSAT2 are immunologically cold: they show little immune-cell infiltration, consistent with HSAT2 actively suppressing anti-tumour immunity [13].

Critically, cells that take up HSAT2-containing EVs do not merely respond and stop: they produce and release their own HSAT2-carrying EVs, which spread to additional cells. This creates a self-sustaining chain reaction. Once triggered, immune suppression continues even if the original cancer signal is removed [3].

3 Extension to Post-Acute Infection Syndromes

We propose that this same chain of events may operate in post-acute infection syndromes (PAIS)—most notably Long COVID and ME/CFS. These conditions frequently begin after a

viral illness, and many patients experience reactivation of herpesviruses that had lain dormant (HSV-1, the cold-sore virus; HCMV, cytomegalovirus; HHV-6).

The proposed sequence is as follows:

Trigger. When a herpesvirus reactivates, it replicates at the centromeres—the pinch points of chromosomes—where viral proteins cause the local DNA to amplify abnormally [4]. This physical disruption destabilises the adjacent pericentromeric regions and switches on HSAT2 transcription [5, 6].

Cumulative erosion. Each viral reactivation damages chromatin structure and strips methylation marks that normally keep HSAT2 off. After repeated or closely spaced infections, the loss of these silencing marks can cross a threshold: HSAT2 RNA production becomes self-sustaining, and the loss of chemical silencing (epigenetic erosion) continues independently of further viral stimulation [1, 7–10]. DNA methylation patterns are known to be altered in ME/CFS and Long COVID, and pericentromeric repeats are especially susceptible to methylation loss.

Propagation. By analogy with cancer, HSAT2 RNAs from cells experiencing centromere disruption are released into the bloodstream in EVs, reach immature myeloid cells, drive MDSC expansion, and induce T-cell exhaustion—contributing to the NK-cell dysfunction and broader immune dysregulation documented in ME/CFS and Long COVID [3]. Because EV-mediated propagation is self-sustaining, immune exhaustion could persist long after the triggering infection has cleared.

4 Testable Predictions

The model generates four predictions testable in patient samples:

- (1) **EV HSAT2 elevation.** HSAT2 RNA or DNA in blood-derived EVs will be elevated in a subgroup of ME/CFS and Long COVID patients with a confirmed viral trigger.
- (2) **Patient subgroups.** Patients who test positive for EV HSAT2 will show a distinct clinical profile: viral-onset history, expanded MDSC populations, and impaired NK-cell function.
- (3) **Biomarker panels.** A combined panel (EV HSAT2, MDSC frequency, cytokines) will identify the affected subgroup with high specificity.
- (4) **Therapeutic response.** Methyl donors—compounds that supply the raw material for DNA methylation (such as SAME [S-adenosylmethionine], methylfolate, and betaine)—may reduce HSAT2 levels and improve symptoms. Conversely, drugs that remove methylation marks (HDAC inhibitors, 5-azacitidine) should be avoided.

These predictions can be pursued immediately. Two HSAT2 detection assays validated in cancer (TRAP-ddPCR for serum RNA and hybridisation capture for plasma cell-free DNA) are ready for use [11, 12]. The simplest starting point is RT-qPCR (reverse-transcription quantitative PCR) on total RNA from patient white blood cells (PBMCs) and serum EVs, stratified by viral trigger. Existing ME/CFS EV sequencing data, such as the total RNA-seq from Seifert et

al. (2026) [14], could be re-analysed for HSAT2 without collecting new samples. Finding elevated HSAT2 in EVs from patients with documented herpesvirus reactivation, co-varying with MDSC expansion, would support the hypothesis; absence in a well-powered cohort would argue strongly against it.

Acknowledgments

We thank Claire Vourc’h (IAB, Université Grenoble Alpes) for expertise on HSAT3—a related pericentromeric repeat—and for initiating a task force to test the HSAT2/ME hypothesis, and Martina Seifert (Charité, Berlin) for discussion of extracellular vesicle sequencing data. This hypothesis was developed by the authors. Within a comprehensive AI-assisted analysis of the ME/CFS literature [15], Opus (Anthropic) evaluated the hypothesis and identified strong mechanistic coherence and cross-disease corroboration. This work builds on two decades of research on repeat sequences as drivers of genome organisation. Preprints [2, 7] are identified as such.

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